AUTONOMOUS MARKET RESEARCH LOOP

MarketMind AI

Next-Generation Multi-Agent Engine for Autonomous Market Intelligence, Fact Verification & Automated Strategic Reports

System Architecture

A resilient multi-agent graph with specialized roles, human-in-the-loop checkpoints, and explicit evidence verification.

Multi-Agent Pipeline Overview

- **Request Decomposition:** Request Analyser parses business goals into structured JSON schemas.
- **Dynamic DAG Planning:** Planner Agent structures non-linear research task graphs.
- **Parallel Execution:** Researcher agents execute API dispatches and live web searches simultaneously.
- **Strict Verification:** Evidence Analyst filters hallucinations and rates source reliability.

Agent Micro-Specialization

Planning & Synthesis Core

Planner Agent: Generates atomic sub-queries and handles dynamic re-planning based on incoming evidence gaps.

Synthesis Agent: Integrates cross-functional findings into cohesive executive summaries with strict source attribution.

Quality & Governance

Quality Control: Runs automated validation loops against completeness rubrics and triggers retries when needed.

CLI Gate: Enables human-in-the-loop approval before running high-cost or external execution tasks.

Core Operating Primitives

Bounded Action Space

Agents execute through standardized schemas (`tools.py`) ensuring deterministic tool inputs and output parsing.

Evidence Verification

Every claim requires structured proof artifacts (`evidence.py`) with source reliability ratings before acceptance.

Autonomous Loop

Iterative feedback loop continuously identifies missing research branches and re-dispatches tasks until quality threshold is met.

Automated Fact Check

Eliminating Hallucinations

MarketMind AI separates information retrieval from report synthesis. All numeric figures, market shares, and trends are independently validated by the Evidence Analyst.

Gaps trigger target tool dispatches until full evidentiary support is achieved.

Research Acceleration

10x

Faster Turnaround Time

From Days to Minutes

Traditional market intelligence requires 15–30 hours of human research, synthesis, and formatting. MarketMind AI completes comprehensive multi-source competitive landscapes in under 5 minutes with zero manual supervision.

Feature & Capability Matrix

Capability	Manual Research	Standard LLM Chat	MarketMind AI
Multi-Source Synthesis	Slow (Hours)	Superficial	✓ Fully Automated
Fact Verification Engine	Human Oversight	✗ Hallucination Prone	✓ Evidence Analyst Gate
Human-in-the-Loop Gate	N/A	✗ Absent	✓ CLI Gate Control
Structured JSON Output	Manual Formatting	Inconsistent	✓ Native Schemas

Benchmark Accuracy & Precision

MarketMind AI achieves superior fact accuracy by combining iterative agent verification loops with structured schema constraints.

Research Execution Lifecycle

1. Ingestion

Request Analyser extracts scope, objectives & constraints.

2. Orchestration

Planner builds task DAG & dispatches Researcher agents.

3. Verification

Evidence Analyst extracts facts & Quality Control validates score.

4. Delivery

Report Generator exports executive PDF/Markdown deliverables.

Enterprise Ready & Scalable

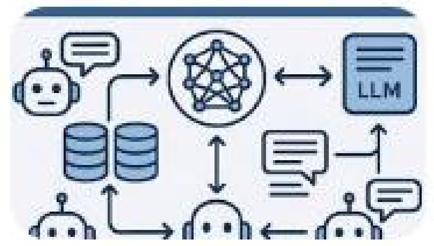
Built with modular state persistence, configurable LLM backends, and robust CLI gates for secure enterprise market intelligence operations.

Questions & Discussion

Explore the MarketMind AI framework and multi-agent implementation.

github.com/marketmind-ai | docs.marketmind.ai

Image Sources



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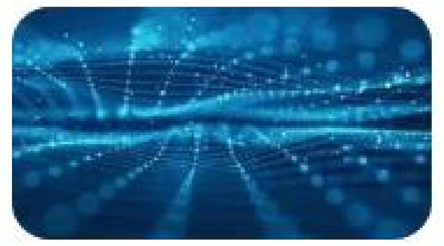

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